Quantification: 543_3
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	11
interior lacuna count	10
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 10)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2218.44	2174	555.066	px^2
major_axis_length_px	lacuna shape	81.5294	80.8669	9.8489	px
minor_axis_length_px	lacuna shape	37.3118	35.2412	10.5528	px
aspect_ratio	lacuna shape	2.2853	2.2947	0.5009	unitless
eccentricity	lacuna shape	0.8782	0.9	0.0743	unitless
solidity	lacuna shape	0.8919	0.9295	0.0895	unitless
roots_count	headline	6.4	5	4.0056	count
ring_length_r30_px	headline	328.1	358	142.164	px
ring_length_r60_px	reported	919.8	1040	363.514	px
owned_length_px	ownership-dependent	3359.93	3403.67	1849.74	px
edge_count	ownership-dependent	155.3	155	85.4141	count
mean_edge_length_px	ownership-dependent	22.2075	21.3181	3.3096	px
perimeter_px	normalised	216.655	219.693	74.8607	px
ring_area_r30_px2	normalised	8553.6	8576.5	1974.23	px^2
ring_area_r60_px2	normalised	21737.1	21875	4704.23	px^2
in_frame_fraction_r30	normalised	0.9906	1	0.0298	unitless
in_frame_fraction_r60	normalised	0.969	1	0.0853	unitless
ring_density_r30	normalised	0.0385402	0.0438411	0.0136468	px^-1
ring_density_r60	normalised	0.0424351	0.0478463	0.0137574	px^-1
roots_per_100px_perimeter	normalised	3.1688	2.7564	1.8226	per 100 px
ring_attached_length_r30_px	network v2	296.5	332.5	136.915	px
ring_attached_length_r60_px	network v2	784.3	938.5	371.642	px
ring_length_w_r30_px	network v2	371.873	404.791	164.612	px
ring_length_w_r60_px	network v2	1046.86	1198.64	415.101	px
sholl_crossings_r10	network v2	8.8	9	4.5412	count
sholl_crossings_r20	network v2	11.1	12.5	4.1753	count
sholl_crossings_r30	network v2	13.9	15	5.1088	count
loop_count	loops, report only	18.4	16.5	12.4561	count
loop_edge_share	loops, report only	0.402	0.4222	0.1556	share
ring_width_mean_r30_px	width	7.54	7.601	0.5987	px

Per field
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measure	value	units
analysed_area_px2	1.02792e+06	px^2
total_skeleton_length_px	36529	px
canalicular_length_density_per_px	0.0355369	px^-1
junction_count	929	count
junction_density_per_px2	0.00090377	px^-2
lacuna_count	11	count

lacunae_per_px2	1.07e-05	px^-2
skeleton_component_count	159	count
mean_component_length_px	229.742	px
median_component_length_px	19	px
field_length_density_w_per_px	0.0404559	px^-1
field_density_without_flagged_per_px	0.0355369	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	224	px
unattached_component_count	15	count
canalicular_length_density_attached_per_px	0.0355369	px^-1
field_length_density_with_unattached_per_px	0.0357548	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	209	count
loop_edge_share	0.4266	share
edge_count	1894	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.7881	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10.198	px
width_pixels_used	36529	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2174	80.8669	35.2412	2.2947	0.9	0.9518	0.4109	79.3965	270.898	False
2	3447	90.3677	63.9486	1.4131	0.7066	0.674	0.4223	197.059	916.902	False
3	1883	69.2617	34.8694	1.9863	0.864	0.9448	0.5742	250.833	340.47	False
4	2052	69.2015	39.4228	1.7554	0.8219	0.9112	0.0622	388.671	369.939	False
5	2266	88.1613	35.5691	2.4786	0.915	0.9021	0.1324	473.05	745.191	False
6	1625	80.1248	27.6985	2.8927	0.9383	0.8325	0.2321	545.572	877.076	False
7	1683	71.0123	31.3735	2.2634	0.8971	0.9355	-0.0518	634.68	198.315	False
8	2203	91.2092	31.4053	2.9043	0.9389	0.9295	0.0832	734.301	635.44	False
9	2633	93.5592	36.2778	2.579	0.9218	0.9454	0.0457	780.576	481.495	False
10	531	28.7881	24.8252	1.1596	0.5063	0.9332	0.5942	1012.28	184.834	True
11								211.745	24.6149	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	7	336	1050	1380.05	63	21.9056	7.4363
2	15	572	1426	4508.26	221	20.3994	7.2258
3	8	380	1030	6004.39	272	22.075	8.3846
4	2	125	344	1147.13	56	20.4845	6.427
5	11	406	1127	3925.78	171	22.9578	8.1001
6	5	426	1281	4505.04	226	19.9338	6.8746

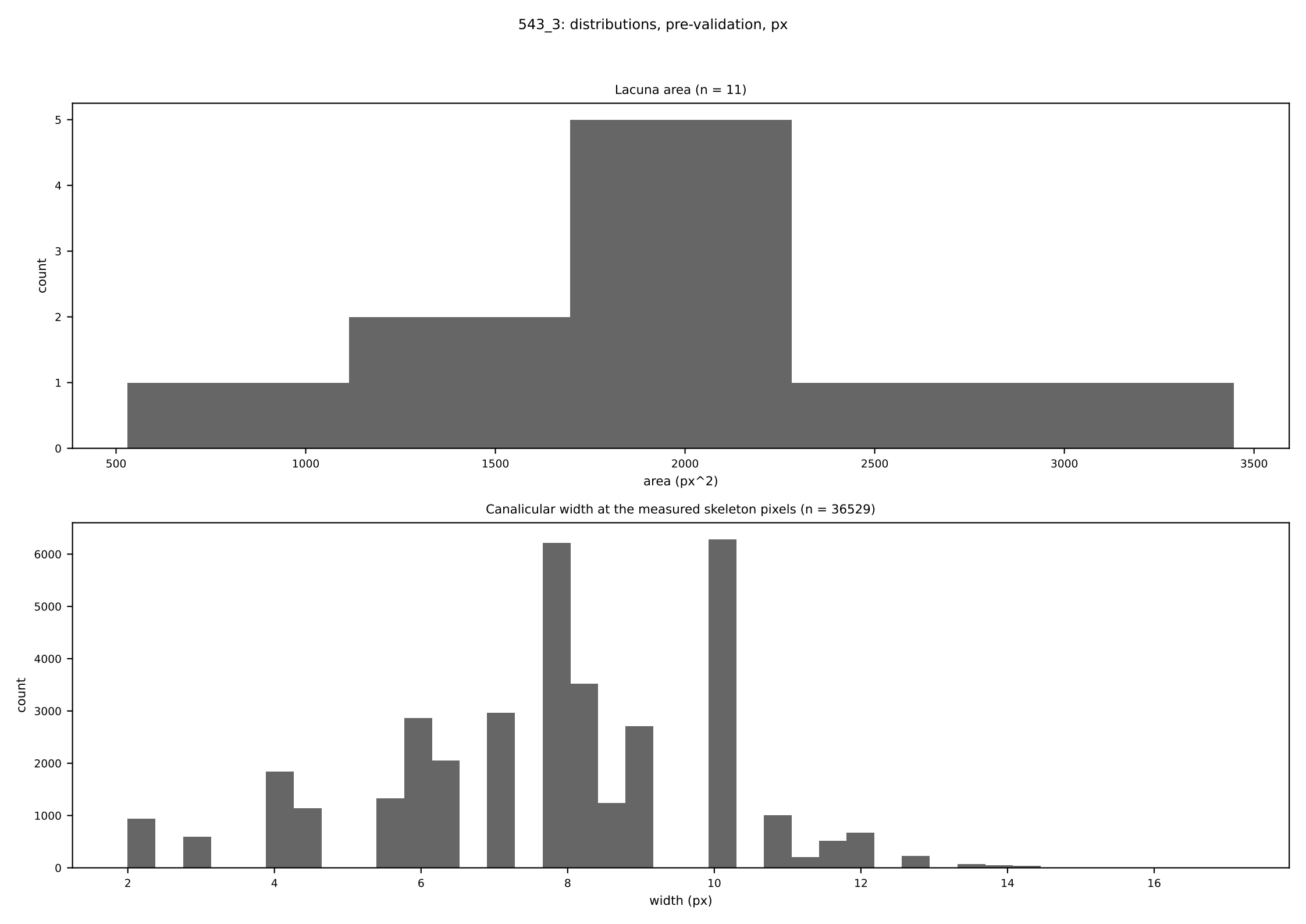
7	3	124	466	2881.56	139	20.7306	8.0362
8	5	409	1115	5774.27	248	23.2834	7.7657
9	4	289	820	1142.89	37	30.889	7.8079
10	2	136	334	872.009	42	20.7621	6.1456
11	4	214	539	2329.91	120	19.4159	7.3422

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	210.066	8595	21715	1	0.9616	0.0390925	0.0483537	3.3323
2	363.446	12097	29071	1	1	0.0472844	0.0490523	4.1272
3	184.794	7941	19515	1	1	0.0478529	0.0527799	4.3291
4	205.965	8333	20507	1	1	0.0150006	0.0167748	0.971
5	241.907	9380	23807	1	1	0.0432836	0.047339	4.5472
6	229.321	8558	22035	1	1	0.049778	0.0581348	2.1804
7	191.965	7943	21525	1	1	0.0156112	0.0216492	1.5628
8	238.108	9212	24077	1	1	0.0443986	0.0463098	2.0999
9	242.936	9386	24420	1	1	0.0307905	0.033579	1.6465
10	95.3909	3354	9525	0.6102	0.5725	0.0405486	0.0350656	2.0966
11	58.0416	4091	10699	0.9057	0.7287	0.05231	0.0503785	6.8916

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	307	905	385.433	1221.61	8	13	17
2	528	1231	659.156	1624.61	18	16	19
3	358	1014	424.149	1175.66	10	12	18
4	107	166	135.983	391.948	3	5	6
5	394	972	452.877	1275.63	10	13	14
6	371	1180	490.617	1457.24	11	15	20
7	96	272	138.154	521.576	3	5	6
8	376	979	462.191	1254.86	12	14	16
9	222	600	331.321	931.98	8	12	13
10	119	302	152.225	375.664	4	4	6
11	206	524	238.853	613.487	5	6	10


543_3: distributions, pre-validation, px
Lacuna area (n = 11)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 36529)
count
width (px)